IS222 – Database Management Systems

Semester I, 2023

Mode: Face to Face & Online (Only Online Submission Required)

Assignment 2 - Category: Group Assignment [Max of. 4 Members Only]

Due Date: Wednesday, 17th May 2023, 11.55 PM (Fiji Standard Time)

Weight: 10%

This assignment tests the following course learning outcomes and the associated CBOK.

CLO 3: Use SQL for data definition, manipulation, and control.

Assignment Objectives

This assignment aims to help you learn Structured Query Language and database design.
This assignment tests your SQL skills. You will work with MySql and write various queries concerning the Data Definition Language (DDL) and the Data Manipulation Language (DML).

***Note:** The attached marking rubric on pages 7-9 will be used to assess this assignment.

Prerequisites

Each student is expected to complete this assignment in groups of four only.

Before you commence this assignment, you should have at least completed the following:

- Completed all required Moodle (Course) activities from Weeks 8 to 11.
- You should have covered all materials and lab exercises from weeks 8 to 11.

*Note: The assignment will primarily examine the content covered from Weeks 8 to 11.

Assignment Overview

A Case Study: Bula & Welcome To Fiji Airways

Fiji Airways is the flag carrier airline of Fiji and operates international services from its hubs in Fiji to 13 countries and 26 cities including, Australia, New Zealand, Samoa, Tonga, Tuvalu, Kiribati, Vanuatu and Solomon Islands, Hong Kong, Japan, Singapore, Canada and the United States. As per your past concrete experiences you are again approached by Fiji Airways (or FA) to assist in designing their new airline database system. They urgently need to have their database set up to keep track of a multitude of information such as commercial aircraft, flights, reservations, etc. FA's CEO, sir, A. Viljoen, and its leading employees provides with the following details: FA owns several aircraft which fly in and out of the various airports worldwide. The Table 1, below is a sample of one of the paper records at FA showing the aircraft's various airport landing sites and dates. Only a few data records are disclosed in the table below from FA. The table below also includes other details of airports and aircraft, which are pivotal and must be recorded in your new database system.

The Aircraft serial number is distinct for each aircraft, and the aircraft's manufacturing dates also need to be tracked. More importantly, there are only certain aircraft types that FA can own due to specific civil aviation regulations. These exclusively include Boeing, Airbus, and Mcdonell Douglas. Civil aviation regulations may provide greater liberty in the future regarding aircraft ownership. However, there is no indication of such as of yet.

Table 1. Sample Details Of FA Aircraft Of Various Airport Landing Sites And Dates.

Airport Code	Airport Name	Airpor t City	Airp ort State	Airport Country	Aircraft Serial Number	Aircraft Model Number	Aircraft Seat Capa City	Aircraft Manufacturing Company	Aircraft Landing Date at Airport	Type of Aircraft
LHR	London Heathrow Airport	London	-	UK	51-11012	777	300	Bach Aircraft	12/2/2015	Boeing
LAX	Los Angeles Internat ional Airport	Los Angeles	California	USA	71-31222	737	150	Malmo	4/8/2015	Boeing
SYD	Sydney Kingsfor d Smith International Airport	Sydney	New South Wales	Australia	71-31222	737	150	Malmo	14/9/2015	Boeing
LAX	Los Angeles Internat ional Airport	Los Angeles	California	USA	11-98281	A380	555	TBM	3/11/2014	Airbus

Furthermore, it is essential to note that specific aircraft (or airplanes) are not permitted to land at certain airports. Additionally, each airport has a specific code, as given in the table above, to distinguish it from the myriad number of airports in the world. Of course, depending on the size and location of the airport, it approves at least some types of aircraft to enter and land. More importantly, FA deals with a large number of passengers and flights in the Pacific. Before booking any flight at FA, a passenger must be registered (if not already registered). The paper-based passenger registration form used by FA is shown below. It provides the passenger details that must be recorded before passengers book flights with FA.

FIJI AIRWAYS PASSENGER REGISTRATION

Date: _____

PASSENGER INFORMATION

____ _____ _____ _____Date of Birth: / /

Title Last Name - First Name
Year/Month/Day

Street Address
& Suburb: _____
_____

City: _____ Prov/State _____ Postal Code/Zip: _____

Country _____

Email _____
Phone Number *(Country Code, Area Code, Number)* ( ) ( ) ( )

Figure 1.

In addition, information about pilots and flights needs to be tracked. Each pilot has a unique 12 character Civil Aviation Authority (or CAA) license number that allows him to operate as a commercial pilot.

The flights operating at FA are essentially non-stop flights going directly from a source location to a final destination point. Hence, no flight legs or segments would constitute stopovers for any given flight. Flight information that must be recorded consists of the specific flight number, e.g., FA931 or FB001, its fare (e.g., $3400.29), number of available seats (e.g., 200), and source and destination details. The destination details are essentially the same as the source details, except they provide the flight arrival details instead of the departure details. An example of the source details for flight number FA931, consists of the source or departing airport, London Heathrow Airport, with a departure date of 2 April, 2014 and a departure time of 0700 (or 7: AM).

Furthermore, each pilot is assigned to none or many flights, while each has precisely one pilot. Of course, each flight corresponds to a particular aircraft, while a plane can be set to one or more flights. Similarly, passengers need to book flights. Each passenger makes at least one flight booking, while each booking corresponds to exactly one passenger. Likewise, each flight can have at least one booking. Similarly, each booking corresponds to a single flight, e.g. FA9102.

Additionally, when a booking is made, certain details must be captured. Some of the core details consist of the booked flight's seat # (ranging simply from 1 to the total number of seats possible in the flight, e.g., 1-200), date of booking (e.g., 12/3/2010), booking time (e.g., 0900 or 9: AM), 12-digit credit card number (of the passenger making the booking), the total cost of the booking given in 2 decimal places e.g., 992.24 and the payment status of the booking. The payment status is a 'Y' or 'N' to represent a Yes or No, respectively, since the passenger can pay for a booking later. Currently, FA allows a passenger to only book one seat on the flight's aircraft within a single booking. If additional seats are required, FA requires that the passenger make another booking. After a booking transaction is complete or a booking is removed if not paid for within a time limit, the number of available seats for the flight will be adjusted. Of course, the total number of seats possible in a flight is the same as the flight's aircraft seat capacity. Consequently, the booking automatically represents the passenger ticket once paid for or remains a temporary reservation if not paid for. Each booking corresponds to a specific booking class, such as Business, Economy, or Executive. The booking class determines the particular meal and entertainment plans, including the type of seat allocated to passengers during their flights. Currently, FA doesn't need detailed tracking of these plans and seats. Instead, a particular booking class defines an additional flight cost factor that is multiplied by the flight fare to get the total booking cost. For instance, a Business class booking for flight FA249, with a flight fare of $2,345.23, would yield an additional flight cost factor of 1.46 (typically recorded to 2 decimal places). This factor is multiplied by the flight fare to get the total booking cost of $3424.04 (rounded to 2. d.p) for the passenger, Chris Bale, flying via this flight. Different booking classes may be added every quarter, and the existing categories may have their cost factors adjusted accordingly. This generally happens on a fortnightly basis. In another example, a booking class, like Business class, would apply to bookings made by passengers: Tim Robbins, Alex Hill, and Harry Pendleton.

This is all the information that has been disclosed by FA thus far. At least the essential requirements must be captured in your database design and eventual implementation.

A thorough potential ERD is shown in Figure 1 below to help you comprehend the preceding scenario better:

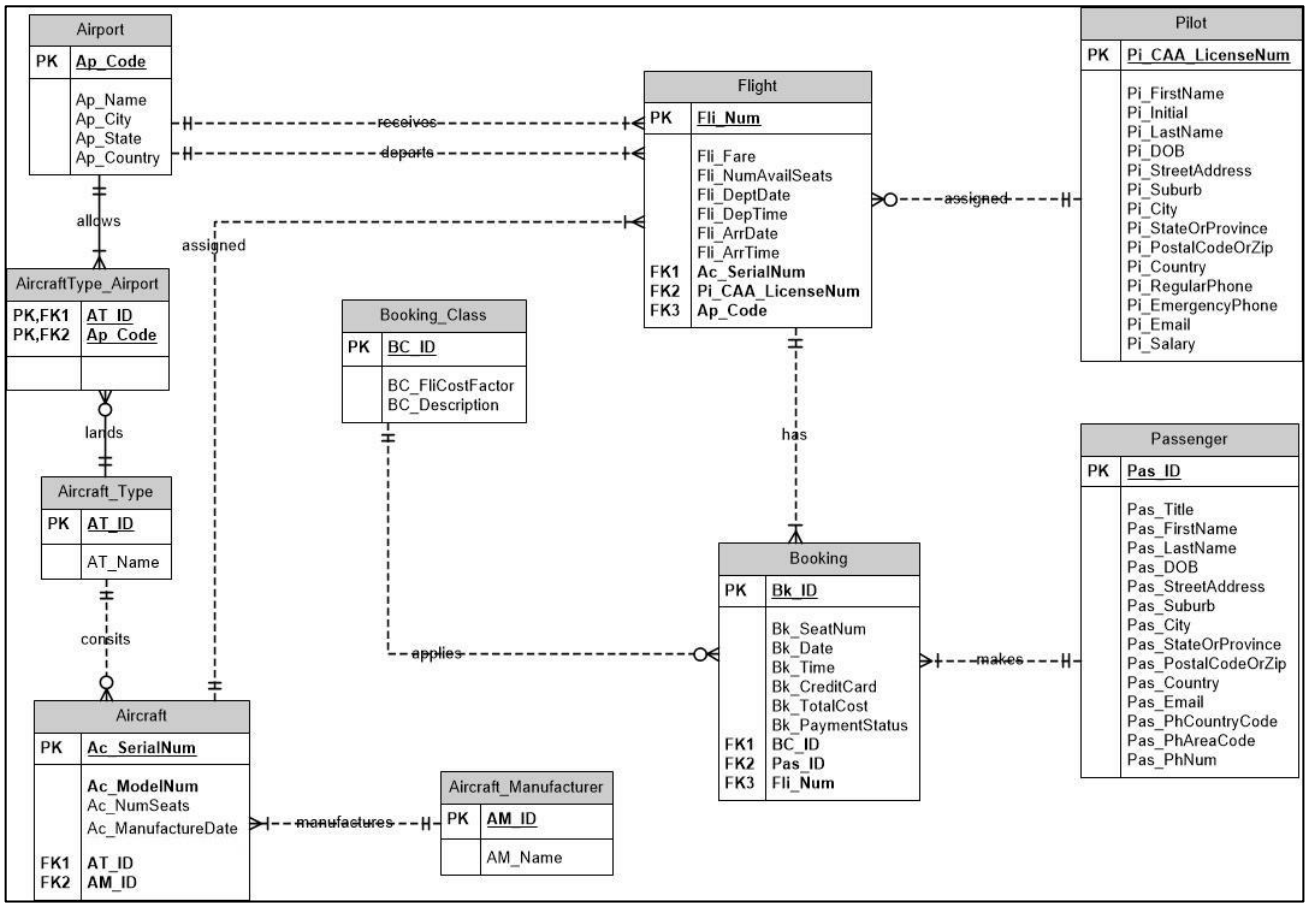

Figure 2.

Following the collation of the requirements,

- You would carry out the conceptual design to create a conceptual schema for the database.
- You would use the above ERD to confirm on the conceptual design. After the conceptual design, we have the logical-design phase to map the high-level conceptual schema onto the implementation data model of the database system that is to be used.
- The last step is the physical design phase, during which the internal storage structures, file organizations, access paths and physical design parameters for the database files are specified.

In the logical design, the implementation data model is typically the relational (SQL) model, and this step basically consists of mapping the conceptual schema defined using the entity-relationship model into a relation schema. In our case, we will carry out this next step of database design of the actual implementation of the database using MySql.
You will need to use the **ERD as the basis for your database implementation. Ensure to use your own records at least 10-15.**

Part A – Creating Table Structures and Inserting Data

You will create the two SQL batch/script files described below with Notepad. We will test these files by running them in the MySql shell. A sample of what your SQL text files could look like is given in DBPart1.sql and DBPart2.sql in Lab 7. Notice how the comments/remarks are used to explain what queries are written using --.

1. Create a Notepad text file called SXXXXXXXX_A2_CreateStructures.sql. The Xs represent your student ID. In this file, you will write all your SQL queries to only generate the table structures as given in the ERD above. This task is part of DDL. Note the following:
    - When you run your script, all the tables should be created with their constraints as given in the ERD above in Figure 2.
    - For the date fields, do not include the time with the date. All times in a day in the database need to be stored in the 24-hour format: 'HH:MM:SS' using the Time data type. Here, HH represents the hours, MM the minutes and SS the seconds. For instance, '22:30:05' would mean 10:30PM with 5 seconds elapsed.
    - Add comments to indicate which SQL statements correspond to which tables using --.
    - Attribute/entity names should be similar if not exactly the same as in the ERD. You can also add any additional useful attributes to the given tables and would not necessarily need to remove any existing ones. Consider any such changes that you introduce carefully.
    - You may add any additional entities that you feel are necessary in the database in accordance with the ERD or above scenario. However, you should not remove the required relationships and entities from the given ERD. Consider any such changes that you introduce carefully.

***** The basic structure in the given ERD must be captured in your implementation in Part A. Any significant changes, such as additions of extra attributes or entities, or any others above, must be

clearly and very briefly explained with comments in the script where the relevant SQL statement is.

- Avoid commands to ignore any particular constraint checks when creating your tables.
- You need the SQL statements to create the tables here. You should be able to create your constraints within the statements to create the tables as well. Any other types of statements may not be needed.

  MUST CHECK: Test out your script in the MySQL shell. When it runs, it is expected to create all the tables with all their required constraints without any errors. Avoid syntax errors.

2. Create another text file in Notepad called SXXXXXXXX_A2_InsertData.sql. The Xs represent your student ID. In this file, you will write all your SQL statements to populate the tables created in the first step with data. This script will only be executed after the first script above. Note the following:

- The data to be inserted in each table is entirely up to you, but should be realistic and pertinent to the scenario. You could use only 10 records per table and follow any constraints given in the scenario.
- Also add comments/remarks to indicate which statements correspond to which tables using --.
- Avoid any code that ignores any particular constraint checks when inserting data.

Part B – Queries

Follow the instructions given below and create a text file called **SXXXXXXXX_A2_Queries.txt.** Write in this file, all the **queries below based strictly on the given ERD**.

- Add comments to indicate the query you are writing, eg. Query 1, Query 2, etc. If the query is ambiguous, you may not get any credit for that query.
- Make sure your queries have no syntax errors. Avoid any gratuitous elements in your queries where possible.Test them out.
- Each query is to generate results based on the requirements given below:

Each query is to generate results based on the requirements given below:

1. **Query 1:** Write a query to display all the details of each booking class without listing each attribute in the query. The results should be sorted in descending order of the flight cost factor.

2. **Query 2:** Write a query to display each passenger's full name (eg. Tom Hanks) aliased as Name, phone number (using the format example: 1-800-4079814, where 1 is the country code, 800 is the area code and the rest of the digits make up the rest of the phone number), including the email of all passengers who have actually purchased less than 10 tickets at FA. Indicate the total number of tickets purchased as well, aliased as Total_Tickets, and sort the results in ascending order of it. Alias the phone number as Phone. Passengers who have not purchased any tickets at all are not required in the result.

3. **Query 3:** List the total number of flights associated with the booking class that is the "Business Class". Alias this as TotalBusinessFlights.

4. **Query 4:** List all the aircrafts that are not allowed to land at the airport with the code, LAX. Just list the aircraft serial number, model number and number of seats. Note that each airport allows at least one aircraft type to land in it and that the database would already have records to reflect this. Sort the results with an explicit ascending order clause on the number of seats.

5. **Query 5:** Write a query to update the pilot salary to double its current figure for all pilots whose current salary is less than or the same as $25,000. The same update should also happen to all pilots that are based in the state of Arizona with a zip code of 85003 in USA (use this abbreviation and not its full form in the query). Only one query is required for this question.

6. Write a query to remove all bookings that have not yet been paid for.

Submission Guidelines:

- Submit the files for parts A and B described below individually on Moodle.
  Do not zip them or combine them as one.
- Submit your files in the Assignment 2 Dropbox by the due date.
  Late submissions will be heavily penalized.
- Any submission made in the original dropbox is not considered late and late submissions take precedence over all prior submissions.
- You may be penalized if you do not follow the submission procedures.
- All names and member IDs are included in one submission from the group.

LATE SUBMISSION = ZERO (0)

Plagiarism

- Incorrect submissions will result in loss of marks or simply a grade of zero being awarded.

- Late Submission = Grade of Zero (0) for Assignment 2.

Zero for any penalty cases thus no two assignment submissions should produce the same results. Any such case will be dealt with severely, with actions including awarding a mark of zero, forwarding the case to the Head of School, and reporting the matter to the Student Disciplinary Committee.

Assignment Queries

- Maintain contact through Moodle discussion forum named Assignment 2 Discussion Forum.

- Feel free to make an appointment via email with the Course Coordinator for any urgent matters regarding Assignment 2.

Assignment 2 Assessment Rubric

Part A Creating Table (Attributes and datatype)	No Tables created *0points*	Few tables created *3points*	More than half the tables created *6points*	All the required tables were created. Some tables have missing attributes *9points*	All tables created with the necessary attributes *12points*
Part A Creating Table (constraints PK and FK)	No tables have any constraints *0points*	Few tables have the necessary constraints *3points*	Half of the tables have the required constraints *6points*	Almost all tables have the required constraints *9points*	All tables have the required constraints *12points*

Part A Inserting Data	No data inserted in any table *0points*	Few tables have some data inserted *3 points*	Half tables have the required inserted data with limited errors. *6points*	Nearly all the tables have some data with no errors *9 Points*	All tables have the required inserted table without errors. *12points*
Part B Query 1	Query do not work and don't make any logic *0points*	Query do not work but the syntax makes some logic *3 points*	Query work but do not produce the required results *5points*	Query works flawlessly with some limitations *7points*	Query works well *10 points*
Part B Query 2	Query do not work and don't make any logic *0points*	Query do not work but the syntax makes some logic *3 points*	Query work but do not produce the required results *5points*	Query works flawlessly with some limitations *7points*	Query works well *10 points*

Part B Query 3	Query do not work and don't make any logic *0points*	Query do not work but the syntax makes some logic *3 points*	Query work but do not produce the required results *5points*	Query works flawlessly with some limitations *7points*	Query works well *10 points*
Part B Query 4	Query do not work and don't make any logic *0points*	Query do not work but the syntax makes some logic *3 points*	Query work but do not produce the required results *5points*	Query works flawlessly with some limitations *7points*	Query works well *10 points*
Part B Query 5	Query do not work and don't make any logic *0points*	Query do not work but the syntax makes some logic *3 points*	Query work but do not produce the required results *5points*	Query works flawlessly with some limitations *7points*	Query works well *10 points*

Part B Query 6	Query do not work and don't make any logic *0points*	Query do not work but the syntax makes some logic *3 points*	Query work but do not produce the required results *5points*	Query works flawlessly with some limitations *7points*	Query works well *10 points*
Submission Guidelines:					All guideline followed well *4 points*

General Comments

The End.